

Black Holes from Hyperbits and from FFGFT

Comparison of two algebraic-geometric approaches — focus: Landauer's principle, bit values, and system-dependent bit energy

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Legend

[K] core result **[B]** algebraically proven **[S]** sketch **[Q]** source **[X]** excluded
[POSTULATE] axiom

Abstract

In “Black Holes from Hyperbits” (IPI 2026) Douglas Matzke derives the entire physics of black holes from the Clifford-algebra axiom $e_i^2 = +1$, without general relativity and without free parameters. Central to this: the mass of a bit follows from Landauer's principle ($E_{\text{bit}} = k_B T_P \ln 2$) plus $E = mc^2$ and is therefore a *universal* number — $m_{\text{bit}} = m_P \ln 2 / (2\pi) \approx 0.11 m_P$, independent of the physical system.

FFGFT (Fundamental Fractal Geometric Field Theory) arrives at a fundamentally different insight: bit values are *system-dependent*. The minimal information unit is the winding quantum $\Delta w = 1$ on T^4 (Doc. 257), and its energy is $E_{\text{bit}} = \hbar c / L$ — dependent on the physical scale L of the system. Landauer's $k_B T \ln 2$ is the special case of one particular scale and temperature, not a universal law (A271, A272, A273, Doc. 302).

This document compares both frameworks and works out the difference in the Landauer interpretation as theoretically essential.

.1 Starting points

Matzke

Single axiom: $e_i^2 = +1$ (hyperbit). **[POSTULATE]**

Physical units via three external bridges: Bekenstein ($4\ell_P^2$ per bit), Compton–Schwarzschild (ℓ_P as length scale), Landauer–Einstein ($m_{\text{bit}} = m_P \ln 2 / (2\pi)$).

FFGFT

Single parameter: $\xi = 4/30000$, derivable as smallest eigenvalue of the spectral operator $\hat{F}$ on $T^4/\mathbb{Z}_3$ **[K]** (Doc. 322). From ξ follow G , ℓ_P , $L_0 = \xi \cdot \ell_P$, $\alpha^{-1} = 137.036$, all lepton masses — without bridge axioms **[K]** (Docs. 180, 011, 006).

Minimal information unit: winding quantum $\Delta w = 1$, energy $E_{\text{bit}} = \hbar c / L$ (scale-dependent) (Doc. 257, A271).

.2 The Landauer question: universal vs. system-dependent bit values

This is the theoretically deepest difference between the two frameworks.

Matzke's position: a universal bit value

Matzke fixes the Landauer energy scale at the Planck temperature T_P :

$$E_{\text{bit}} = k_B T_P \ln 2 \quad \Rightarrow \quad m_{\text{bit}} = \frac{m_P \ln 2}{2\pi} \approx 0.110 m_P \quad (\text{Matzke, Ch. 6})$$

This value is *universal*: every bit carries the same mass, independent of the system (electron, black hole, vacuum). From it follow the stability threshold $n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) = 6.41$ and the Schwarzschild relation directly.

FFGFT position: system-dependent bit values

FFGFT understands Landauer's principle as a statement about *phase-space regions* of a thermal carrier ensemble, not about abstract information (A271):

*Counting proceeds by distinguishability (orthogonality), not by energy. Thermodynamics counts regions and does not read them — there is no thermodynamic truth detector. Sufficient for heat production is the **non-bijectivity of the region map**, not logical irreversibility.*

The fundamental bit energy in FFGFT is (Docs. 257, 302):

$$E_{\text{bit}} = \frac{\hbar c}{L} \quad (\text{FFGFT, Doc. 257})$$

where L is the physical scale of the system. Landauer's $k_B T \ln 2$ is the limiting case at one particular scale and bath temperature (A273: "two floors — the thermal $k_B T$ and the quantum-geometric $\hbar c / L$; the larger one governs").

The bit energy accordingly varies between scales:

System	Scale L	$E_{\text{bit}} = \hbar c/L$
Planck scale	$\ell_P = 1.62 \times 10^{-35} \text{ m}$	$m_P c^2 = 1.22 \times 10^{19} \text{ GeV}$
FFGFT floor	$L_0 = \xi \cdot \ell_P = 2.15 \times 10^{-38} \text{ m}$	$E_0/\xi \approx 56 \text{ PeV}$
Characteristic	$L_e = \hbar c/E_0 \approx 26.85 \text{ fm}$	$E_0 = 7.35 \text{ MeV}$
Nuclear physics	$\sim 1 \text{ fm}$	$\sim 197 \text{ MeV}$
Room-temperature bit	$L_T \sim k_B T/(\hbar c) \sim \mu\text{m}$	$\sim 3 \times 10^{-21} \text{ J} = 18 \text{ meV}$

FFGFT critique of Matzke's Landauer bridge

Matzke sets $T = T_P$ in $E_{\text{bit}} = k_B T \ln 2$. This yields a *universal* bit value. FFGFT objects: Landauer's bound holds for a thermal equilibrium ensemble at bath temperature T — a statement about the *carrier*, not about abstract information (A272). The Planck temperature T_P is not the temperature of the physical system (electron, proton, black hole in the cosmos); it is merely an energy scale. Landauer holds because of mechanics (Liouville), not because of a privileged scale.

Consequence: Matzke's universal stability threshold $n_{\text{thresh}} = 6.41$ hinges essentially on the choice $T = T_P$. With a system-dependent bit energy, n_{thresh} would itself be system-dependent. **[S]**

Two floors, one principle

FFGFT distinguishes (A273, Doc. 290):

Static Bit $\equiv$ winding quantum $\Delta_w = 1$ on T^4 : scale-universal, energy-independent as a counting quantity; the energy $E_{\text{bit}} = \hbar c/L$ is scale-specific.

Thermodynamic Irreversible erasure: $Q \geq k_B T \ln 2$ — cost in joules, dependent on the bath temperature T . This price has nothing to do with the information-theoretic quantity (A272: "no energy can be assigned to an information bit, only to its carrier").

Kinematic Reversible bit-flip time: $\mathfrak{T} = \hbar/E_{\text{bit}}$; Margolus–Levitin $\tau \geq \pi\hbar/(2E)$; energy sets the clock without loss (Doc. 290).

The four guises of a bit energy at the characteristic scale $E_0 = 7.35 \text{ MeV}$:

$$m_0 c^2 = E_0 = k_B T_0 = \hbar/\mathfrak{T} \quad \Rightarrow \quad T_0 = E_0/k_B \approx 8.5 \times 10^{10} \text{ K} \quad (1)$$

Landauer at room temperature gives $Q \approx 3 \times 10^{-21} \text{ J} \approx 18 \text{ meV}$ — eight orders of magnitude below E_0 . The statements "what the bit is" ($E_0 = 7.35 \text{ MeV}$) and "what erasure costs" ($k_B T \ln 2 = 18 \text{ meV}$ at 300 K) belong to different layers (Doc. 290, A271).

.3 Bekenstein entropy and Planck voxels

Matzke

1 bit = $\text{Cl}(6) = 3$ Witt pairs = 1 Planck voxel $4\ell_P^3$. **[B]**

$S_{\text{BH}} = A/(4\ell_P^2) = V_{\text{shell}}/(4\ell_P^3)$; Bekbits as base unit, $\Omega = 2^b$.

FFGFT

Minimal information unit: winding quantum $\Delta_w = 1$ on $\pi_1(T^4) \cong \mathbb{Z}^4$. Discreteness is topological (no half winding), not from a tiling (Doc. 302).

Planck voxels exist implicitly as orbifold fixed points. Bekenstein: $S_{\text{BH}} = A/(4\ell_{\text{P}}^2)$ as a consequence of the Bekenstein bound on the saturation surface **[B]**.
Holographic direction: N -dimensional $\rightarrow$ 3D (as Matzke), not $3D \rightarrow 2D$.

Convergence: fermion generation = algebraic base unit

Matzke: $\text{Cl}(6) = 3$ Witt pairs = Planck voxel = one fermion generation (Furey). **[B]**
FFGFT: $T^4/\mathbb{Z}_3$ orbifold, three $\mathbb{Z}_3$ sectors $\rightarrow \text{SU}(3)_c \rightarrow N_c = 3$ algebraically necessary $\rightarrow$ one fermion generation. **[B]** (Doc. 321)
Both find: threefold structure, algebraically enforced, no free parameter.

Divergence: depth of derivation

Matzke presupposes $\text{Cl}(6)$ as physical unit (**[POSTULATE]** via Furey identification).
FFGFT derives the threefold structure from $\mathbb{Z}_3$ trialization: $N_c = 3$ follows with algebraic necessity **[B]** (Doc. 321, R79–R80).

.4 Schwarzschild radius and Bekenstein entropy

Matzke's Novel Result #1:

$$r_s = \frac{2}{\sqrt{\pi}} \sqrt{S_{\text{BH}}} \ell_{\text{P}} \quad (\text{Matzke})$$

The factor $2/\sqrt{\pi}$ comes from the transition hypercube $\rightarrow$ sphere ("rounding the square").
FFGFT:

$$G = \frac{\xi^2}{4m_e}, \quad \ell_{\text{P}} = \sqrt{\frac{\hbar G}{c^3}}, \quad r_s = \frac{2GM}{c^2} \quad (\text{FFGFT, Doc. 180})$$

Bekenstein: consequence of the bound $S \leq 2\pi RE/(\hbar c)$ on the saturation surface **[B]**.

Numerically: $20 M_{\odot} \Rightarrow r_s = 59.1 \text{ km}$ (both exact).

Convergence

Both derive r_s without the GR field equations. Both show: ℓ_{P} is enforced by self-consistency, not set axiomatically. Numerically identical.

Divergence

Matzke's $2/\sqrt{\pi}$ follows from hypertetrahedron geometry. In FFGFT no such form factor appears; $r_s = 2GM/c^2$ is the standard relation, and G comes from ξ .

.5 Stability threshold

Matzke

$n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) = 6.41$ bits. **[B]**

$\text{Cl}(6)$ fermion: 6.000 bits, 0.41 bits below threshold.

A stable universe, because the minimal fermion algebra lies algebraically below the collapse threshold.

Dependence: n_{thresh} hinges essentially on $m_{\text{bit}} = m_p \ln 2/(2\pi)$, i.e. on the Landauer bridge with $T = T_p$.

FFGFT

Stable matter follows from $\mathbb{F} \cdot m = 1$: lepton masses from rational invariants (r_i, p_i) of the torus space **[K]** (Docs. 006, 046, 320).

No direct bit-counting threshold; stability from the orbifold fixed-point structure.

FFGFT does not know n_{thresh} as a bit count — and it should be system-dependent if bit values are system-dependent.

Open bridge — a system-dependent threshold?

Matzke's $n_{\text{thresh}} = 6.41$ is elegant but hinges on the postulate $m_{\text{bit}} = m_P \ln 2 / (2\pi)$ (universal bit value, Planck temperature). From the FFGFT standpoint the threshold would be system-dependent:

$$n_{\text{thresh}}(L) = \frac{2\pi}{\sqrt{2}} \cdot \frac{m_P c^2}{E_{\text{bit}}(L) \cdot \sqrt{2}} = \frac{2\pi}{\sqrt{2}} \cdot \frac{m_P \cdot L}{\sqrt{2} \hbar / c} \quad (2)$$

For $L = \ell_P$ (Planck): gives the same number as Matzke. For $L = L_0 = \xi \cdot \ell_P$ (FFGFT floor): n_{thresh} would be smaller by a factor ξ .

Question: Is $n_{\text{thresh}} = 6.41$ the right number for the universe, or is it a consequence of the choice $L = \ell_P$? **[S]**

.6 Hawking radiation

Matzke

Hawking quanta are exclusively massless vacuum states $\{V, V_s\}$ of $\text{Cl}(6)$. Annihilation: $P_k^+ \cdot P_-^{\text{vac}} = 0$ (exact in $\mathbb{Z}_3$). **[B]**

Information conserved: each V encodes the orientation of the destroyed Bekenstein voxel.

$T_H = T_P / (4 n_{\text{mass}} \ln 2)$. **[B]**

No formation radiation (formation = time reversal of annihilation).

FFGFT (Doc. 325)

Complete mechanism from FFGFT-native building blocks:

Temperature: KMS rule $T_H = \hbar / (k_B \tau)$ with $\tau = 8\pi G M / c^3$ from $\mathbb{F} \cdot m = 1$ **[K]** (Docs. 313 G, 325).

Quantum: $k_B T_H = \hbar c / (4\pi r_s)$ exactly — the Hawking quantum is the system-dependent bit of the horizon scale **[K]** (Doc. 325).

Selection: only $T_R = 0$ escapes; absorption $P_j P_k = 0$; confinement = Hawking selection **[B]** (Docs. 321, 325).

Carrier: massless $n_4 = 0$ torus mode; massive modes fall back (clock compression) **[K]** (Doc. 325).

Information: sector pair $(k, -k)$ orthogonally readable, unitarity of the spectral operator **[K]** (Docs. 322, 325).

Convergence — two complete mechanisms

Both frameworks possess an algebraically complete mechanism. The correspondences are structurally exact (Doc. 325): Matzke's $P_k^+ \cdot V_s = 0 \leftrightarrow$ FFGFT's $P_j P_k = 0$; Matzke's massless $V \leftrightarrow$ FFGFT's $n_4 = 0$ torus mode; Matzke's voxel orientation $\leftrightarrow$ FFGFT's sector pair $(k, -k)$.

Divergence — the bit-value foundation

Matzke's mechanism rests on the universal bit value $m_{\text{bit}} = m_P \ln 2 / (2\pi)$; the FFGFT route (KMS from $\mathbb{F} \cdot m = 1$ plus system-dependent bit energy) does without it and shows: it is a reparametrization of the KMS relation, not an independent physical input (Doc. 325). In addition FFGFT delivers the power correction $P = P_{\text{std}}(1 - \xi \ln(M/m_P)) = 0.6\text{--}1.4\%$, for which Matzke has no counterpart.

.7 Comparison table

Quantity	Matzke/hyperbit	FFGFT
Fundamental object	$e_i^2 = +1$ [POSTULATE]	$\xi = 4/30000$ from spectrum [K]
Bit energy	$E_{\text{bit}} = k_B T_P \ln 2$ (universal) [POSTULATE]	$E_{\text{bit}} = \hbar c / L$ (system-dependent) [K]
Role of Landauer	physical bridge: $m_{\text{bit}} = m_P \ln 2 / (2\pi)$	carrier operation, not abstract info (A271)
Bit value	universal, $\approx 0.11 m_P$	system-dependent, scale- L -dependent
Planck length	Compton Schwarzschild [B]	$= G = \xi^2 / (4m_e)$, $\ell_P = \sqrt{\hbar G / c^3}$ [K]
Minimal length	ℓ_P	$L_0 = \xi \cdot \ell_P \approx \ell_P / 7500$ [K]
Stability threshold	$n_{\text{thresh}} = 6.41$ bits [B]	orbifold fixed-point structure, no bit count
r_s without GR	$(2/\sqrt{\pi}) \sqrt{S_{\text{BH}}} \ell_P$ [B]	$2GM/c^2$ with G from ξ [K]
α^{-1}	not derived	137.036 from $D_f = 3 - \xi$ [K]
Lepton masses	not derived	m_e, m_μ, m_τ from (r_i, p_i) [K]
$\sin^2 \theta_W$	not derived	0.2308 [K] (Doc. 323)
Hawking mechanism	V/V_s annihilation [B]	KMS + $\mathbb{Z}_3$ selection [K]/[B] (Doc. 325)
$N_c = 3$ algebraically	Furey identification [POSTULATE]	$\mathbb{Z}_3$ trialization [B] (Doc. 321)
Holography	$N\text{-dim} \rightarrow 3\text{D}$ [B]	$T^4/\mathbb{Z}_3$ projection [K]
Information conservation	V encoding of voxel orientation [B]	unitarity of the spectral operator [K]

.8 FFGFT objections to Matzke's Landauer bridge

Three concrete objections from the FFGFT corpus:

1. **Landauer applies to carrier operations, not to abstract information** (A272, A273). What Landauer 1961 proves is a lower bound for *carrier operations on a thermal equilibrium ensemble* — not a universal statement about abstract information. The identification $m_{\text{bit}} = m_P \ln 2 / (2\pi)$ treats the bit as a thermodynamic object at Planck temperature. That is a realization assumption, not a law.

2. **The bit energy is scale-specific** (Docs. 302, 257). In FFGFT the minimal information unit is the winding quantum $\Delta w = 1$ on T^4 — scale-universal as a *counting quantity*, but with scale-specific energy $E_{\text{bit}} = \hbar c/L$. Landauer's $k_B T \ln 2$ is the special case at one particular scale and bath temperature. The phrase “the bit energy is $m_p \ln 2/(2\pi)$ ” implies a scale that is not distinguished in FFGFT.
3. **The stability threshold is not a purely algebraic statement** (derived from points 1 and 2). $n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) = 6.41$ follows from the Compton–Schwarzschild equation plus $m_{\text{bit}} = m_p \ln 2/(2\pi)$. The first piece (Compton = Schwarzschild) is geometric **[B]**. The second (Landauer at T_p) is a postulate. The apparent elegance of the result (Cl(6) exactly 0.41 bits below threshold) hinges on this postulate. A system-dependent bit energy would shift the threshold.

.9 Open bridges

Points with potential for mutual cross-fertilization

- [S]** n_{thresh} **from the orbifold structure**: Doc. 325 shows that Matzke's bit value is a reparametrization of the KMS relation. The constructive counter-question remains open: can the FFGFT fixed-point condition for fermion stability be translated into an effective bit-counting statement — and which system-specific bit energy then enters?
- [S]** **Casimir connection** (Doc. 324, R84): $\xi = C_2(\text{SU}(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4$. This links the algebraic structure of Cl(6) (Casimir invariant $C_2 = 4/3$, from $N_c = 3$ **[B]**) with the geometric scale of FFGFT (torus mode number $N_F = 10^4$). This bridge is closed **[K]** (Doc. 324).
- [S]** $\sin^2 \theta_W$ **from Cl(6)**: FFGFT derives $\sin^2 \theta_W(M_Z) = 0.2308$ **[K]** (Doc. 323). Can Matzke's Cl(6) structure produce an analogous value?

.10 Status balance

Result	Matzke	FFGFT
r_s without GR	[B]	[K]
ℓ_p from self-consistency	[B]	[K]
Cl(6) = fermion generation	[B] (Furey)	[B] (orbifold)
$N_c = 3$ algebraically necessary	[POSTULATE]	[B]
Bit energy system-dependent	[X]	[K]
$\alpha^{-1} = 137.036$	[X]	[K]
Lepton masses	[X]	[K]
$\sin^2 \theta_W$	[X]	[K]
Hawking mechanism worked out	[B]	[K]/[B] (Doc. 325)
n_{thresh} (bit count, universal)	[B]	[X] (rejected: system-dep.)
Information conservation	[B]	[K]
Holography $\neq 3D \rightarrow 2D$	[B]	[K]

Conclusion

Matzke's hyperbit framework and FFGFT are complementary structures with remarkably similar structural findings: fermion generation = algebraic base unit, $N_c = 3$ enforced, Schwarzschild radius without GR.

The deepest difference lies in the treatment of bit values. Matzke posits a universal bit value via the Landauer bridge at Planck temperature — an elegant but, by FFGFT standards, inadmissible universalization. FFGFT holds: bit values are system-dependent ($E_{\text{bit}} = \hbar c/L$), Landauer applies to carrier operations of a thermal ensemble, and the stability threshold n_{thresh} would then not be a universal number.

This is no objection to the algebraic beauty of Matzke's approach, but a theoretical difference in foundations. Doc. 325 makes it constructive: the complete Hawking physics — temperature, selection, carrier, information conservation — follows from $\mathbb{F} \cdot m = 1$, the $\mathbb{Z}_3$ structure and the system-dependent bit energy, without a universal bit value. Matzke's $n_{\text{thresh}} = 6.41$ thus demonstrably hinges on the bit-value parametrization, not on the temperature physics.

Main finding

Matzke and FFGFT converge on **fermion generation = fundamental algebraic unit**. They diverge on **bit values**: universal (Matzke, Landauer at T_P) vs. system-dependent (FFGFT, $E_{\text{bit}} = \hbar c/L$). This difference is not cosmetic — it concerns the stability threshold and the mass of a bit. The Hawking temperature formula itself is untouched by it: both routes give the same number, but only the FFGFT route (Doc. 325) does without a universal bit value.

Relation to the FFGFT corpus

Topic	Status	Doc.
Landauer: phase-space bookkeeping	[K]	A271
Carrier vs. information (Landauer critique)	[K]	A272
Bit energy system-dependent (computing sphere)	[K]	A273
Information unit $\Delta w = 1$, $E_{\text{bit}} = \hbar c/L$	[K]	Doc. 257
Elementary cells, bit capacity, partition invariance	[K]	Doc. 302
$G = \xi^2/(4m_e)$, ℓ_P , L_0	[K]	Doc. 180
$SU(3)_c$ from $\mathbb{Z}_3$ trialization	[B]	Doc. 321
$\sin^2 \theta_W(M_Z) = 0.2308$	[K]	Doc. 323
$\xi = C_2(SU(3))/N_{\text{Fourier}}$	[K]	Doc. 324, R84
Hawking mechanism (KMS, $\mathbb{Z}_3$, $n_4=0$)	[K]/[B]	Doc. 325
Spectral theory, Hilbert-space embedding	[K]	Doc. 322
Lepton masses from (r_i, p_i)	[K]	Docs. 006, 046, 320
$E = mc^2 = E = m$ (unit identity)	[K]	Doc. 077

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